

AIR QUALITY INDEX ANALYSER

This project focuses on analyzing and predicting Air Quality Index (AQI) levels in India, utilizing a consolidated and cleaned dataset of historical daily ambient air quality records from 1990 to 2015. The data was sourced from the Ministry of Environment and Forests and the Central Pollution Control Board of India under the National Data Sharing and Accessibility Policy (NDSAP).

Key components of the project include:

- **Data Analysis:** A detailed examination of air quality across various states and years, with a focus on the concentration of pollutants like sulfur dioxide (SO₂), nitrogen dioxide (NO₂), respirable suspended particulate matter (RSPM), and suspended particulate matter (SPM).
- **AQI Prediction:** Using supervised learning models, the project predicts AQI values based on pollutant concentrations. Algorithms employed include Linear Regression, Logistic Regression, Random Forest Classifier, and Decision Tree Classifier.
- **AQI Classification:** Air quality is classified into categories such as "Good," "Satisfactory," "Moderately Polluted," "Poor," "Very Poor," and "Severe."

THE COLUMNS ARE

- stn_code (station code)
- sampling_te (date of sample collection)
- state (Indian State)
- location (location of sample collection)
- agency

- type (type of area)
- so2 (sulfur dioxide concentration)
- no2 (nitrogen dioxide concentration)
- rspm (respirable suspended particulate matter concentration)
- spm (suspended particulate matter)
- location_monitoring_station
- pm2_5 (particulate matter 2.5)
- Date

CALCULATE AIR QUALITY INDEX FOR SO₂ BASED ON FORMULA

The air quality index is a piecewise linear function of the pollutant concentration. At the boundary between AQI categories, there is a discontinuous jump of one AQI unit. To convert from concentration to AQI this equation is used

$$I = I_{low} + \frac{I_{high} - I_{low}}{C_{high} - C_{low}}(C - C_{low})$$

WHAT IS SULFUR DIOXIDE?

Sulfur dioxide is a gas. It is invisible and has a nasty, sharp smell. It reacts easily with other substances to form harmful compounds, such as sulfuric acid, sulfurous acid and sulfate particles.

About 99% of the sulfur dioxide in air comes from human sources. The main source of sulfur dioxide in the air is industrial activity that processes materials that contain sulfur, e.g. the generation of electricity from coal, oil or gas that contains sulfur. Some mineral ores also contain sulfur, and sulfur dioxide is released when

they are processed. In addition, industrial activities that burn fossil fuels containing sulfur can be important sources of sulfur dioxide.

Sulfur dioxide is also present in motor vehicle emissions, as the result of fuel combustion. In the past, motor vehicle exhaust was an important, but not the main, source of sulfur dioxide in air. However, this is no longer the case.

CONCLUSION

- AQI is highly correlated with all the independent variables(so₂, no₂, rspm, spm)
- AQI has been increasing over the years.

Best models for AQI range classification :

1. Random Forest Classifier
2. Decision Tree Classifier